Name.: _____ **Class:** _____

BLANTYRE SECONDARY SCHOOL CLUSTER
2018 MSCE MOCK EXAMINATION

MATHEMATICS

Subject Number: M131/II

Friday, 23 March **Time allowed: 2 h 30 mins**
8:30 – 11:30 am

PAPER II
(100 marks)

Instructions
1. This paper contains -- pages.
2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Calculators may be used.
5. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.
6. **All working must be clearly shown;** it should be done on the same sheet as the rest of the answers.
7. Write your **name** on top of each page.

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SECTION A (55 Marks)
Answer all the six questions.
1. a. Given that
$f(y) = 2xy - 35y^2$ *and* $g(x) = 2x^2 + xy$.

Factorise completely $f(y) + g(x)$

(5 marks)

b. Solve the equation
$3x^2 = 7x + 2$, giving your answer correct to 2 decimal places.

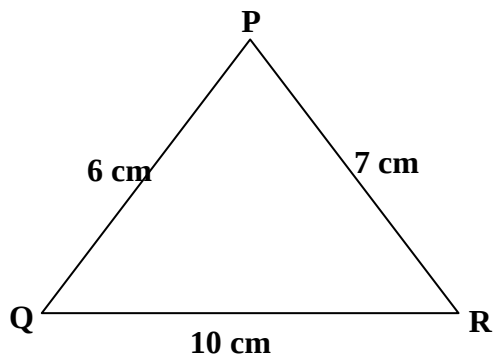
(7 marks)

2. a. Given that $P = \begin{bmatrix} 3 & 4 \\ 1 & -2 \end{bmatrix}, Q = \begin{bmatrix} 1 & 3 \\ 6 & 1 \end{bmatrix}, R = \begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix}$. *Find* $3\,(Q - PR)$

Continued/. . .

(6 marks)

b. In **Figure 1**, **PQR** is a triangle such that **PQ =** 6 cm, **QA** = 10 cm and **RP** = 7 cm.



Calculate angle **PRQ** to the nearest degree

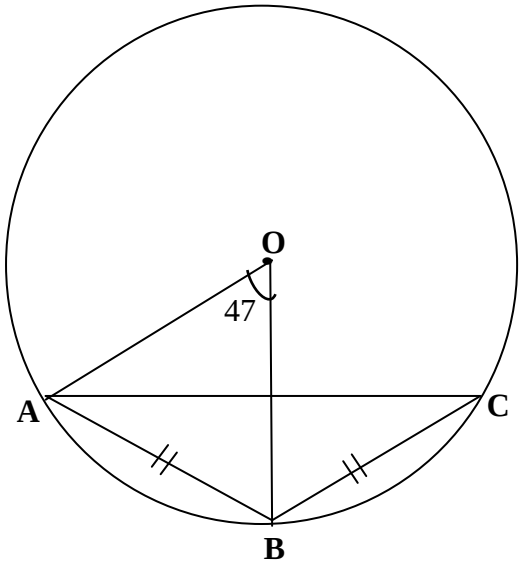
(5 marks)

3. a. Solve the equation $3^{2h+1} - 28\,(3^{h-1}) + 1 = 0$

Continued/. . .

(8 marks)

4. a. **Figure 2** is a circle with centre **O** and **AOB** and **ABC** are triangles in which **AB = BC**. Angle **AOB =** 74 ˚.



Calculate **OAC.**

(5 marks)

b. Simplify $\frac{v-a}{v+a} - \frac{v+a}{v-a}$.

Continued/. . .

(5 marks)

5. a. Given that $C^1 = (5 – 5)$ is the image of C (1,1) after translation. Find the translation vector that maps C to C^1.

(3 marks)

b. Find the standard deviation for a set of data whose deviation from the mean are -2, -3, 4, 1, leaving your answer to 3 significant figures.

(5 marks)

6. The third term of GP is $\frac{4}{9}$, the fifth term is $\frac{16}{81}$ and common ratio is positive. Find the common ratio and the first term.

Continued/. . .

(6 marks)

SECTION B

Choose any **three** questions from this section.

7. a. Solve the equation

$$2x^3 - 5x^2 + x + 2 = 0$$

(10 marks)

b. Simplify $\frac{3\sqrt{2}+1}{\sqrt{3}+\sqrt{6}}$ with a rational denominator.

(5 marks)

8. a. (i) Copy and complete the table for $y = (x + 1)(x^2 + x - 6)$.

X	- 4	- 3	- 2	- 1	0	1	2	3
y	- 18	0	4		- 6	- 8		24

Continued/. . .

(ii) Use the scale of 2 cm to represent 1 unit on x-axis and 5 units on y-axis, draw the graph for $y = (x + 1)(x^2 + x - 6)$.

(ii) Use your graph to solve the simultaneous equation;

$$y = x^3 + 2x^2 - 5x - 6$$
$$y = x - 4$$

(10 marks)

(b) The angle of depression of a car from the top of a pole is 35 °. If the top of the pole is 21 m from the ground, calculate the distance of the car from the pole.

9. a. Solve the simultaneous equation

$$x + 3y - 1 = 0$$
$$x^2 - 8y^2 - y + 5 = 0$$

b. Make R subject of the formula

$$B = N\left(1 + \frac{R}{X}\right)^K$$

Continued/. . .

10. An object starts from rest and accelerates uniformly at $5m/s^2$ in 4 seconds. It further accelerates uniformly to a velocity of 90m/s in the next 3 seconds. It maintains this velocity for 2 seconds and then it comes to rest in another 5 seconds.

(i) Using a scale of 2 cm to represent 20m/s on the vertical axis and 2 cm to represent 2 seconds on the horizontal axis, draw a speed-time graph for the motion of the object.

(ii) Using your graph, calculate the acceleration in the last 5 seconds. (8 marks)

Continued/. . .

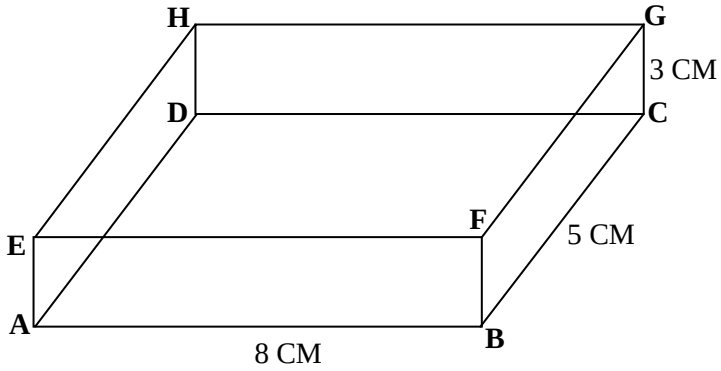
b. A family has three children born at different times. Assuming it is equally likely to have a baby boy or a girl;
(i) Draw a tree diagram to show the probability of having a boy or a girl on each of the three births.

(ii) Calculate the probability the family has 2 boys and 1 girl in any order of births.

(7 marks)

11. a. There are 60 vitamin pills which contain at least one of the following vitamins; **A, B, C**. 12 pills contain vitamin **A** only, 7 pills contain vitamin **B** only, 11 pills contain vitamin **C** only, 6 pills contain all the vitamins. $\cap (B \cap C \cap A^{1}) = x$, $\cap (A \cap C \cap B^{1}) = 2x$, $\cap (A \cap B \cap C^{1}) = 3x$.
Use a venn diagram to calculate the number of pills which contain vitamin **A**.

b. **Figure 3** shows a water tank with rectangular faces; **AB** = 8 CM, **BC** = 5cm, **CG** = 3 cm.



Calculate the angle which **AG** makes with the plane **ABCD**.

(7 marks)

12. a. The cost of a meal per person at a hotel is partly constant and partly varies inversely as the number of people. When there are 3 people the cost of a meal is K1600. When there are 9

Continued/. . .

people, the cost of a meal is K1200. What will be the cost of a meal per person if there are 20 people?

(7 marks)

b. Mala has K6000 to buy skirts and blouses. Skirts cost K600 each while blouses cost K300 each. She wants to spend at least K1200 more on skirts than on blouses. She would like to buy at least 4 skirts and 2 blouses.

(i) If x represents number of skirts and y represent the number of blouses, write down four inequalities in x and y.

(ii) Using the scale of 2 cm to represent 2 units on both axes, draw a graph to show the region which represents the four inequalities.

Continued/. . .

(8 marks)

END OF QUESTION PAPER
NB: This paper has 13 pages